

Guozhen Zhu, Ph.D.

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Education

Ph.D. in Electrical Engineering, University of Maryland, College Park Aug 2023
Thesis: [Wireless Sensing and Analytics for Motion Monitoring and Mapping](#) | Advisor: Dr. K. J. Ray Liu
M.S. in Electrical Engineering, University of Maryland, College Park Aug 2022

Summary & Research Focus

AI Research Scientist with **8+ years** of experience driving ML research into **million-scale deployment**. Focused on **foundation-model pretraining** and **representation learning** for multimodal/temporal perception, **robustness and generalization** in real-world settings, and **LLM/agent methods** for embodied/spatial reasoning and decision-making.

Experience

Principal Scientist, Origin Wireless, Inc. Sept 2023 – Present

- Built a **foundation model** for ambient perception via large-scale pretraining, achieving **strong cross-task transfer** across presence, fall detection, localization, and motion understanding.
- Studied **downstream multitask adaptation** of the pretrained backbone, improving **data efficiency** and performance consistency across target tasks.
- Built a large-scale text-grounded ambient perception dataset and benchmarked baseline multimodal alignment methods.
- Built a large-scale, in-the-wild **benchmark dataset** enabling representation learning under deployment constraints and hardware heterogeneity.
- Invented **TruShield**, an interpretable and privacy-preserving system for through-wall human presence detection; deployed at scale and recognized with the **CES 2025 TWICE Picks Award**.
- Led the evolution of a deployed ambient-intelligence solution on **over 25 million devices** from statistical methods to learning-based models.
- Advanced multimodal spatial intelligence across wireless, inertial, and device-free sensing systems, achieving robust real-world generalization under heterogeneous hardware conditions.
- Translated research into products deployed on **over 25 million telecom routers** and **over 100 million smart devices** through collaborations with global partners.

Graduate Research Assistant, University of Maryland, College Park Sept 2019 – May 2021

- Developed learning-based perception pipelines for motion understanding, mapping, and localization from high-dimensional sensor data.
- Proposed signal-informed learning methods to improve generalization in noisy real-world environments.
- Designed a multimodal mapping and tracking system integrating wireless and robotic sensing, reducing data requirements by **88%**.
- Reviewer for leading venues: ICML, IEEE TMC, IEEE IoTJ, ACM TIOT, and ACM IMWUT.

Selected Publications

- **Guozhen Zhu**, Yuqian Hu, Weihang Gao, Wei-Hsiang Wang, Beibei Wang, K. J. Ray Liu. *AM-FM: A Foundation Model for Ambient Intelligence Through WiFi*. Forty-Third International Conference on Machine Learning (ICML), 2026, submitted.
- **Guozhen Zhu**, Yuqian Hu, Weihang Gao, Wei-Hsiang Wang, Beibei Wang, K. J. Ray Liu. *CSI-Bench: A Large-Scale In-the-Wild Dataset for Multi-task WiFi Sensing*. 2025 39th Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.
- **Guozhen Zhu**, Yuqian Hu, Wei-Hsiang Wang, Beibei Wang, K. J. Ray Liu. *Experience Paper: Scaling WiFi Sensing to Millions of Commodity Devices for Ubiquitous Home Monitoring*. ArXiv preprint arXiv:2506.04322, 2025.
- **Guozhen Zhu**, Beibei Wang, Weihang Gao, Yuqian Hu, Chenshu Wu, K. J. Ray Liu. *SrcSense: Robust WiFi-based Motion Source Recognition via Signal-Informed Deep Learning*. IEEE Journal of Selected Areas in Sensors, 2024.
- Sai Deepika Regani, Beibei Wang, Yuqian Hu, **Guozhen Zhu**, K. J. Ray Liu. *FallAware: An Explainable Learning Approach to Robust Fall Detection With WiFi*. IEEE Journal of Selected Areas in Sensors, 2024.
- Yuqian Hu, **Guozhen Zhu**, Wenhao Wang, Beibei Wang, K. J. Ray Liu. *What you need is a good CSI*. Proceedings of the 30th Annual International Conference on Mobile Computing, 2024.
- WeiHsiang Wang, Beibei Wang, Yuqian Hu, **Guozhen Zhu**, K. J. Ray Liu. *Device-Free Room-Level Localization With WiFi Utilizing Spatial-Frequency-Time Diversity*. IEEE Internet of Things Journal, 2024.

- **Guozhen Zhu**, Beibei Wang, Yuqian Hu, K. J. Ray Liu. *WiFi-Based Robust Human and Non-human Motion Recognition With Deep Learning*. 2024 IEEE International Conference on Pervasive Computing and Communications.
- Yuqian Hu, **Guozhen Zhu**, Wenhao Wang, Beibei Wang, K. J. Ray Liu. *Robust Proximity Detection Using On-Device Gait Monitoring*. 2023 IEEE 9th World Forum on Internet of Things.
- **Guozhen Zhu**, Beibei Wang, Yuqian Hu, Chenshu Wu, Xiaolu Zeng, K. J. Ray Liu. *Wi-MoID: Human and Nonhuman Motion Discrimination Using WiFi With Edge Computing*. IEEE Internet of Things Journal, 2023.
- **Guozhen Zhu**, Chenshu Wu, Beibei Wang, K. J. Ray Liu. *EZMap: Boosting Automatic Floor Plan Construction With High-Precision Robotic Tracking*. IEEE Internet of Things Journal, 2022.
- **Guozhen Zhu**, Chenshu Wu, Xiaolu Zeng, Beibei Wang, K. J. Ray Liu. *Who Moved My Cheese? Human and Non-human Motion Recognition with WiFi*. 2022 IEEE 19th International Conference on Mobile Ad Hoc and Smart Systems.
- **Guozhen Zhu**, Chenshu Wu, Beibei Wang, K. J. Ray Liu. *Floor Plan Reconstruction with High-Precision Rf-Based Tracking*. 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).
- **Guozhen Zhu**, Bin Yan, Mengting Xing, Chunna Tian. *Automated Counting of Bacterial Colonies on Agar Plates Based on Images Captured at Near-Infrared Light*. Journal of Microbiological Methods, 2018, 153:66-73.

Selected Patents

- **Guozhen Zhu**, Yuqian Hu, Beibei Wang, Muhammed Zahid Ozturk, Sakila Jayaweera, Wei-Hsiang Wang, Weihang Gao, Jiaxuan Zhang, K. J. Ray Liu, Oscar C. L. Au. *Wireless sensing using a foundation model*. U.S. Patent App. No. US20250337657A1.
- **Guozhen Zhu**, Beibei Wang, Weihang Gao, Yuqian Hu, Chenshu Wu, Xiaolu Zeng, K. J. Ray Liu, Oscar C. L. Au. *Method, apparatus, and system for wireless sensing based on deep learning*. U.S. Patent No. US12352889B2. (Granted 2025)
- **Guozhen Zhu**, Chenshu Wu, Beibei Wang, K. J. Ray Liu, Oscar C. L. Au. *Method, apparatus, and system for map reconstruction based on wireless tracking*. U.S. Patent No. US12078510B2. (Granted 2024)
- Yuqian Hu, **Guozhen Zhu**, Beibei Wang, K. J. Ray Liu, Oscar C. L. Au. *Method, apparatus, and system for wireless proximity detection*. U.S. Patent No. US12256360B2. (Granted 2025)
- Sakila S. Jayaweera, Beibei Wang, **Guozhen Zhu**, Weihang Gao, Chenshu Wu, K. J. Ray Liu, Oscar C. L. Au. *Wireless sensing for in-vehicle child presence detection*. U.S. Patent No. US12352844B2. (Granted 2025)
- Beibei Wang, Muhammed Zahid Ozturk, Chenshu Wu, Xiaolu Zeng, Sai Deepika Regani, Yuqian Hu, K. J. Ray Liu, Oscar C. L. Au, Yi Han, Hung-Quoc Duc Lai, David Claffey, Chun-I Chen, Dan Bugos, Peng Zan, **Guozhen Zhu**. *System and method for radio-assisted sound sensing*. EU Patent EP4102247B1. (Granted 2024)
- Beibei Wang, Chenshu Wu, **Guozhen Zhu**, Muhammed Zahid Ozturk, Fengyu Wang, K. J. Ray Liu, Hung-Quoc Duc Lai, Sakila Sandeepani Jayaweera Samaranayake Arachchige Dona, Sai Deepika Regani, Xiaolu Zeng, Wei-Hsiang Wang, Oscar C. L. Au, Min Wu. *System and method for radio-based sleep monitoring*. EU Patent App. No. 22200097.8.
- **Guozhen Zhu**, Beibei Wang, Chenshu Wu, Xiaolu Zeng, K. J. Ray Liu, Oscar C. L. Au. *Method, apparatus, and system for wireless human and non-human motion detection*. U.S. Patent App. No. 18/391,529.
- **Guozhen Zhu**, Beibei Wang, Yuqian Hu, Xiaolu Zeng, K. J. Ray Liu, Weihang Gao, Oscar C. L. Au, Chenshu Wu. *System and method for wireless human-nonhuman motion classification*. EU Patent App. No. 24150597.3.
- Wei-Hsiang Wang, Beibei Wang, **Guozhen Zhu**, K. J. Ray Liu, Oscar C. L. Au, et al. *Wireless sensing using measurement enhancement*. U.S. Patent App. No. US20250141715A1.
- Sakila S. Jayaweera, Sai Deepika Regani, **Guozhen Zhu**, Wei-Hsiang Wang, Beibei Wang, K. J. Ray Liu, Oscar C. L. Au, et al. *High resolution human imaging using neural network*. U.S. Patent App. No. 19/004,301.

Industry Impact

Technology deployed worldwide across multiple sectors:

- **Verizon Home Awareness**: Ambient intelligence deployed on **millions** of Fios routers nationwide (based on US Patent No. US12256360B2 and App. No. 18/391,529).
- **Signify (Philips) & Nami Sensing**: On-device perception deployed on **more than 100 million** smart lights and plugs (based on US Patent No. US12256360B2, and App. No. 18/391,529).
- **TruShield Security**: Award-winning privacy-preserving human detection system (**CES 2025 TWICE Picks Award**, based on US Patent No. US12352889B2 and App. No. 18/391,529).
- **Verisure WiFi Vision & Deutsche Telekom**: EU deployment with security operator and Internet Service Providers (based on EU Patent App. No. 24150597.3 and US Patent App. No. 18/391,529).
- **Arlo Technologies**: Exclusive partnership for Verified Human Presence detection (based on US Patent No. US12352889B2 and App. No. 18/391,529).